

Fake News Detection for COVID-19 using PySpark Framework MLlib

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Abstract— Besides causing a very severe health crisis for the whole world, the COVID-19 epidemic has speeded the spread of fake news and false information. Solving false news problems is important in these time-sensitive conditions to provide correct information and guarantee people's safety. This work aims to use PySpark, a distributed data processing platform, to create a prediction model for identifying COVID-19 fake news. The experiments are conducted on a dataset sourced from Kaggle and a prediction model is created using PySpark's machine learning libraries, such as MLlib and SparkML, to classify news articles as real or false. Different classification methods, such as logistic regression, decision trees, and random forests, are researched and improved for the best performance. The accuracy, precision, recall, and F1 score of the model are assessed using relevant measures, confirming its efficacy in detecting bogus news. The best results were achieved by the logistic regression classifier which achieved accuracy and F-measure of 0.9.

Keywords: COVID-19, Fake news, PySpark, MLlib, Logistic regression, Decision trees, Random Forest

I. INTRODUCTION

The global environment witnessed the unprecedented meeting of digital communication and news consumption caused by the outbreak of COVID-19 [1,2]. The emergence of a paradigm where people are the main news sources for social media platforms is the convergence. The digital era has brought a time of instant information gratification when stories and related information can spread, be discussed and are amplified over digital platforms. Real-time interactions, engagement of the users and viral content are what social media platforms give priority to by design which is the reason for the evolution. The COVID-19 pandemic [3,4] is a particularly strong illustration of this transformation. The rapidity with which the virus spreads is the same with which information is spread over social media, be it true or false. News reports, positions and rumours were blown up by the world and the choices of policies as well as public perceptions and behavioral responses were affected. This elevation in news casts serious doubts about the value, accuracy, and truthfulness of the content carried by digital media.

Social media is a popular source of news due to its distinctive benefits. The fact that social media platforms disseminate news articles in a real-time manner automates the delayed time connected with traditional news organizations, social media provides information delivery that is more immediate than traditional forms of journalism. News narratives can reach a larger audience not having so much distribution costs because news dissemination is highly cost-

effective in digital transmissions. Possibly, the most important feature of social media is the participatory nature of the platforms which transforms reading the news from being a set-in-stone activity into a communal one. An environment whereby information suppliers and consumers have a mutually advantageous relationship through readers' capability to share, comment and submit their content. While there are disadvantages, there are the same significant drawbacks. The clear divide between true and false information has further become uncertain, by the fact that news has become democratized. Digital means of production and distribution of news are quite affordable and accessible, yet it allows for fake news, which is information that deliberately tries to deceive people, to be easily sent across the media platform. Such content has been occasioned by financial and political gains that entities make through spreading propaganda, eroding public trust in information sources and the society is put into disarray.

Advanced approaches combining technology like deep learning and transform-based models [5,6] can also be used for information verifications that are very much needed to deal with the complex ecosystem. The research takes the COVID-19 pandemic as an essential case study and applies PySpark which is an advanced distributed data processing environment that will be used for making a scalable and robust model.

PySpark [7,8] is a Python API of Apache Spark distributed computing system that allows for large-scale data processing. It brings Python as a part of Spark's capabilities, making Spark more convenient and user-friendly. Many kinds of data processing tasks such as batch, stream, machine learning, and graph processing can be done using PySpark [9]. Due to its ability to do large-scale data processing over a distributed network, it has become a go-to for big data analytics. For banking, health care, and e-commerce, handling of data cleaning, analysis, and advanced machine learning models are translated into a simpler task. Adaptability, scalability and Python's package interoperability are all the reasons for its prominent role in large data processing and analytics.

The proposed method is an efficient approach utilizing the PySpark and advanced machine learning techniques which is very instrumental in creating a predictive model that correctly identifies the misinformation about the pandemic. The core objectives of this model are to ensure the consistent classification of the news as either real or fake. The study in question greatly enriches the discussion on the reach of information disbursement and uses logistics regression to counter the complex and time-sensitive problem of COVID-19 fake news.

The remaining paper is organized as follows: Section II introduces the literature work related to this paper, Section III discusses the proposed methodology, Section IV presents the results of the proposed model, Section V concludes the findings of the paper and Section VI gives all the references that are included in the paper.

The key contributions of the paper are:

1. Contributes an optimum machine learning model to classify fake news.
2. The proposed model uses MLlib of PySpark framework which would contribute to a scalable and robust model for fake news detection.

II. RELATED WORKS

There have been numerous studies on fake news conducted so far. Everyone has been concerned lately due to the rise of bogus news. Several scholarly articles regarding false news in several languages have previously been published. Many researchers have employed deep learning algorithms and machine learning algorithms to identify false information.

Shaikh, et al. [10] suggested 3 ML algorithms and listed extraction approaches that are used as solutions to fake news identification. humanizing the given sentence, we have 84.05%, 95.05%, and 92.9% respectively. Khanam Z, et al. [11] aimed to detect fake news in this research by analyzing it at two levels, disclosing and characterizing. In the first stage, the main ideas of fake news are shared on the social media platform which eventually creates false news. We considered a couple of supervised learning models, XGBoost, Random Forest, and Decision Tree, to detect fake news during the discovery stage. A suggestion by Halder was to add the total words, the total unique words, type/token ratio, and the number of sentences in order to improve the accuracy of the Naive Bayes algorithm. Babar and team [12] presented propositions of hybrid models of N-gram language modelling and LSTM integration. The proposed model indicates the results that are striking in precision, recall rate and computational time; thus, helping to create a more aesthetic method for detecting fake news. Baria and team [13] provided a method for discovering the best features and procedures for identifying fake news. To choose a set of characteristics as a criterion in detecting false news. They developed and implemented a system through which news data is preprocessed, cleaned, steamed, N-gram encoded, a bag of words, and TF-IDF.

Mayank Kumar, et al. [14] proposed a text-based fake news detection mode using Basic ML classifiers such as Random Forest (RF), Naive Bayes (NB), Support vector machine (SVM), Logistic regression, and KNN, apart from this BOW TF and BOW TF-IDF vectors were used reduction methods lemmatization, stemming, and chi-square test on vocabulary. Al Asaad et al. [15] tried to evaluate news stories' reliability in this paper by analyzing their features. They discovered that, across both classifiers, the model's BoW and TF-IDF yielded the best accuracy scores. Huxiao et al. [16]

proposers suggested an FNDMS, a framework for identifying false news, in this research. To assess the veracity of a news event, FNDMS employs a multi-source scoring technique that integrates the credibility scores of many news sources.

Torabi Asr et al. [17] proposed a model that uses the MLlib of PySpark framework for implementing the model for fake news detection. The PySpark framework contributed to a scalable and robust model for fake news detection.

The survey showcased deep learning and machine learning algorithms experimented with for fake news detection. However, very few studies have previously worked on Big Data frameworks for fake news detection. In this research we are using PySpark Big data framework to identify the fake news articles to stop the spread of fake news.

III. PROPOSED METHODOLOGY

A. Description of the Dataset:

The corpus of the COVID-19 dataset is made up of 7000 news articles, and this corpus serves as the dataset used in the study. Every item of news, by itself, is labelled as either real or fake; this is the point of departure for the development of a prediction model to sort and ascertain fake news since it is extremely important for understanding the spread of disinformation in the pandemic. The "News" field is a journalistically obtained text from different sources that covers a variety of topics relating to COVID-19, including virus transmission, prevention measures, treatment choices, government responses, and also public health guidelines. The diversity of the text in this dimension is very well represented by the multitude of existing information coupled with exaggerations, lies, or misrepresentations. The label field is the baseline by which the dataset is evaluated, assigning a label to each article as true or fake news. Binary classification permits easy distinguishing of real news articles, which contain information from credible sources, and fake news articles, which spread confusion by publishing misleading or deceptive content. The suggested approach for identifying fake news is shown in Fig. 1 and is covered in more detail in the subsections that follow.

B. Data Pre-processing

To process the raw corpus of the textual data, the pre-processing steps are devised. Normalization of the text is achieved using a technique that involves changing every character in a text to lowercase so that the capitalization is consistent and the same throughout the text. This technique makes text mining more exact as one can do more intense comparisons and computations, after which all inconsistencies are eliminated and take away the hesitation of stepping into the text analysis. Letter case normalization helps the process and analyze text effortlessly and accurately for both the computational and also human users because no longer do the variations in letter case cause dissimilarities in the results of text-based operations.

¹ <https://www.kaggle.com/datasets/csmalarkodi/covid-fake-news-dataset>

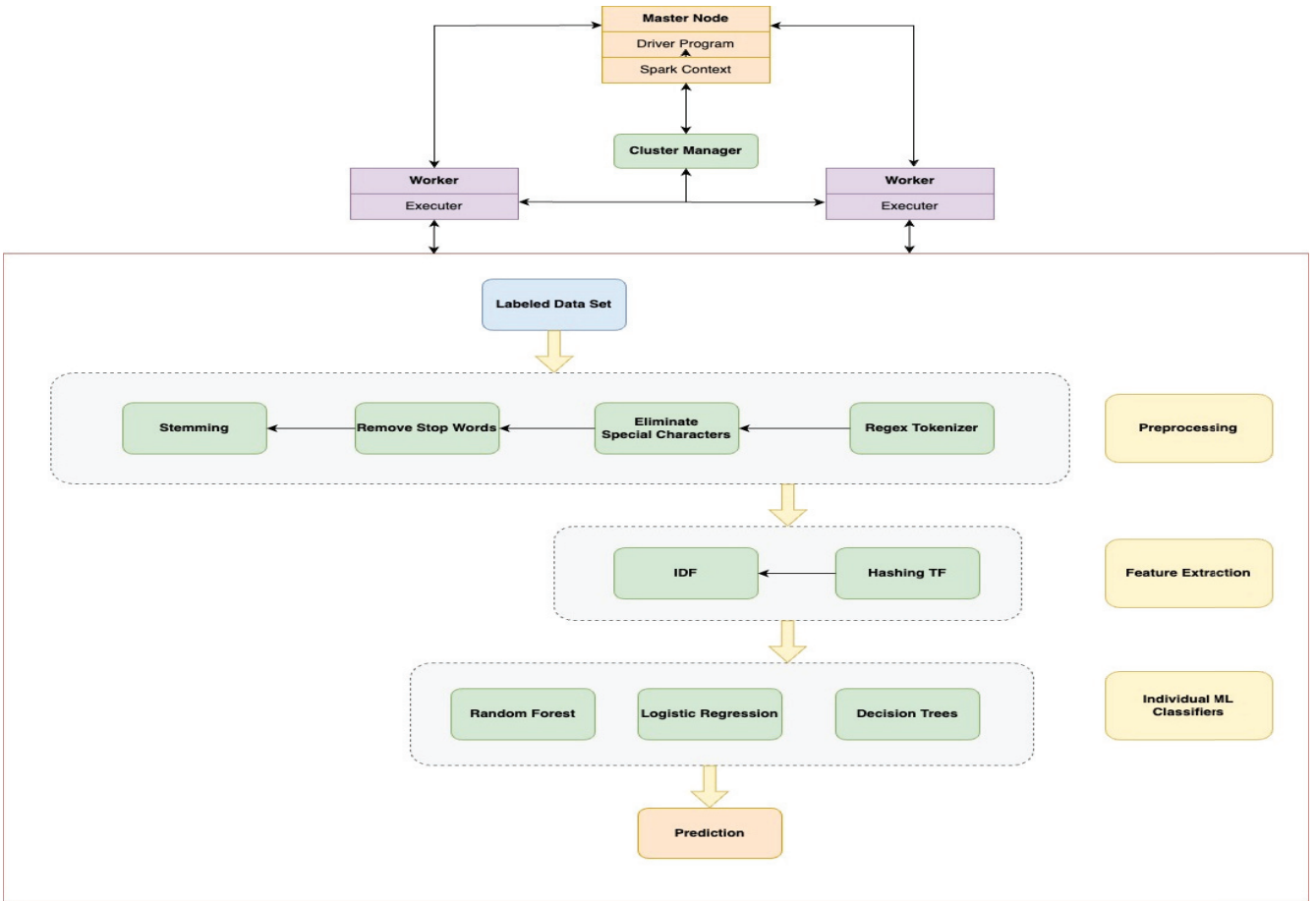


Fig. 1. Proposed Methodology for Fake News Detection

a) *Regex Tokenization*: Text tokenization, a process of simultaneously breaking down pre-processed text into separate units, i.e., either a single word or phrase, is one of the most prevalent steps to prepare the text data for the machine learning algorithms. In the study that we create, we apply a regular expression tokenizer to extract and differentiate the important words and phrases, which are subsequently in a format that the machine learning models can use to process the sentences.

b) *Eliminate Special Characters*: Eliminating letters, punctuation, and symbols that are not needed in the text to make it semantic is the most important part of the text data prepping. By deleting unnecessary material, this method enhances the quality and clarity of the texts through elimination of the superfluous elements. The process of URL removal consists of the removal of any possible web addresses or links that may be present in any text. The purpose of removing the URLs from the text is to keep it relevant and strictly analyzable. These links normally do not contain any comprehensible content both for analysis or understanding.

c) *Remove Stop words*: Among the most important aspects of text preprocessing is the removal of stop words, words such as 'and', 'the', and 'is' that are widespread and have little semantic value. Through this process, the text data becomes better for effectiveness, usefulness, and dimensionality reduction for the next stage of analysis. The overused terms are thus removed, specifying this way more educational material and advancing language comprehension accuracy. This makes the machine learning algorithms dealing with text more efficient.

d) *Stemming*: Stemming is a text processing technique that engages in implementing algorithms to break down words into their basic root words, enabling the words that are related to be sorted out. The goal of that strategy is to speed up the text analysis by mapping different words to a single basic dictionary set. For example, stemming would lead to both forms of “running” and “ran” to the same root, “run.” Making the word forms appear equal, this reduction process enhances text comprehension, information retrieval, and a lot of natural language processing activities and this ultimately brings better understanding and correction of language and language use.

C. Feature Extraction

a) *Hashing Term Frequency*: Hashing TF creates a sparse vector space by turning the tokenized words into numerical vectors. The frequency of each phrase inside a given feature vector is represented using this technique. To capture the core of the text while reducing computational complexity, the number of features is set at a sufficiently high value.

b) *Inverse Document Frequency*: The feature vectors are further refined using the Inverse Document Frequency (IDF) component. IDF emphasizes unique phrases while underscoring generic ones. IDF increases the model's ability to distinguish between different classes of data by giving terms weights based on how unique they are within the overall dataset.

The core of this research is the thorough evaluation of categorization models, each defined by distinctive characteristics and skills, to identify the most appropriate model for the job. Three primary classification techniques are considered:

D. Machine Learning Classifiers

Following pre-processing and feature extraction, the data is fed into machine learning classifiers such as Random Forest, Decision Tree, and Logistic Regression.

a) *Logistic Regression*: Logistic regression, the standard model, is well known for its clarity, interpretability, and efficiency in binary classification applications. It is the perfect place to begin the inquiry because of its intuitive character.

b) *Decision Trees*: In order to classify data, decision trees offer a non-linear method that can capture complex decision boundaries. These trees provide a way to model intricate relationships in the data.

c) *Random Forests*: An ensemble method called random forests makes use of the combined strength of various decision trees. Random forests improve predictive accuracy while reducing overfitting by combining forecasts from several trees.

Each classification model is carefully analyzed, looking at its advantages and disadvantages. The study aims to determine, based on the selected evaluation parameters, the most efficient method for identifying fake news connected to COVID-19. The chosen model must consistently exhibit high levels of accuracy, precision, recall, and F1, which indicate that it can distinguish between fake news and real news pieces.

IV. RESULTS

The performance of the trained machine learning models, which comprise Random Forest, logistic regression, and decision tree classifiers, is evaluated; the results are shown in Table 1. The graphical representation of the F-measures reported by the models is also presented in Fig. 2. With an accuracy rate of 91%, logistic regression demonstrated its superior performance for the task on COVID-19 fake news identification using PySpark. On the other hand, the accuracy of decision trees and random forests was 76% and 77%, respectively. Because of its dominance, logistic regression is the most dependable method for separating fact from fiction throughout the pandemic. Moreover, the precision and recall values of logistic regression are balanced over both the classes unlike random forest and decision tree classifiers which exhibit an imbalance across classes. The ROC curves for all three classifiers are presented in Fig.3, Fig.4 and Fig.5 respectively. The area under the ROC curve measurement which is an additional metric to the accuracy metrics attests to the fact that logistic regression performs better than the decision trees and also random forests in fake news detection in COVID-19. The area under the curve for logistic regression remains very stable at a reported value of 0.9. This is saying how logistic regression produces an optimal trade-off between the true positive and false positive rates even at different decision threshold levels.

TABLE I. MODEL PERFORMANCE OF ML CLASSIFIERS

Model	Precision		Recall		F1- Score		Accuracy
	Fake	Real	Fake	Real	Fake	Real	
<i>Logistic Regression</i>	0.88	0.93	0.92	0.89	0.90	0.91	0.91
<i>Random Forest</i>	0.85	0.73	0.61	0.91	0.71	0.81	0.77
<i>Decision Tree</i>	0.75	0.76	0.70	0.81	0.73	0.78	0.76

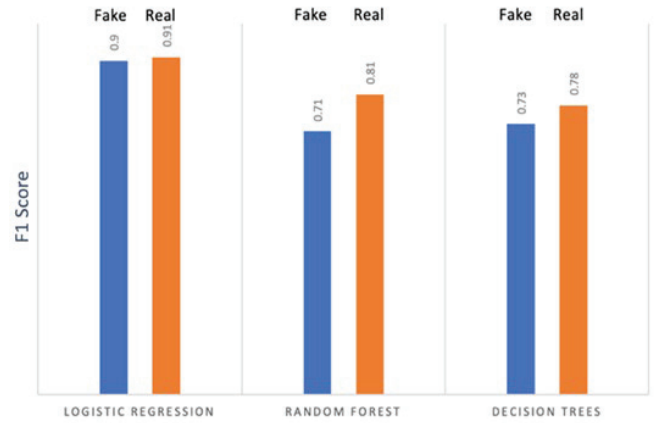


Fig. 2. F1 score comparison of ML models

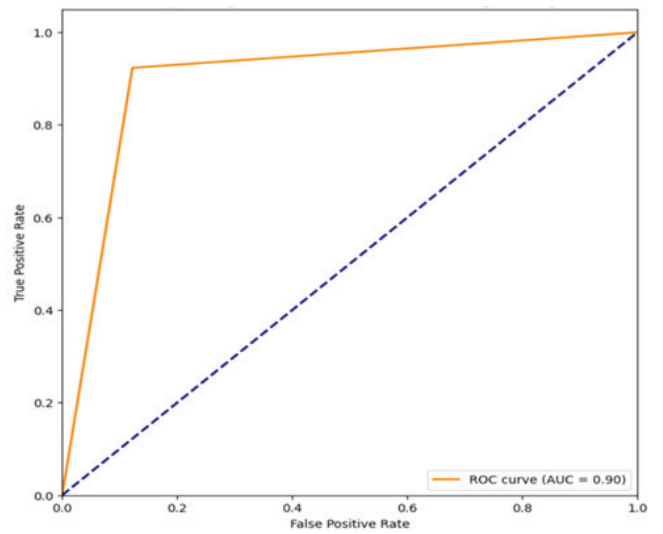


Fig. 3. Logistic Regression classifier ROC curve

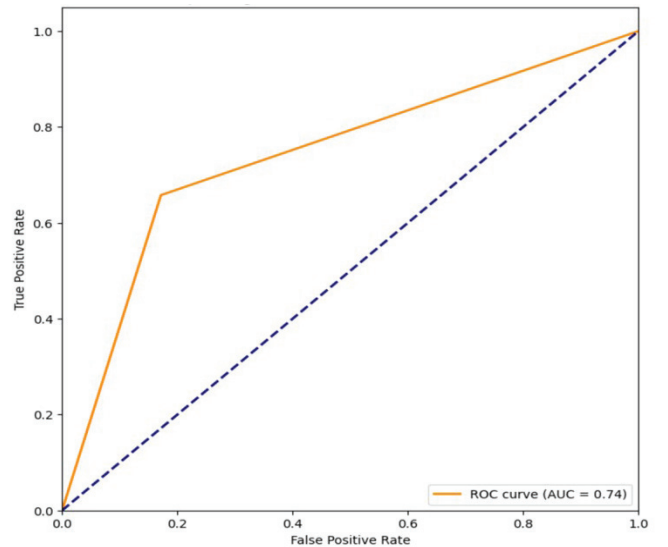


Fig. 4. Random Forest classifier ROC curve

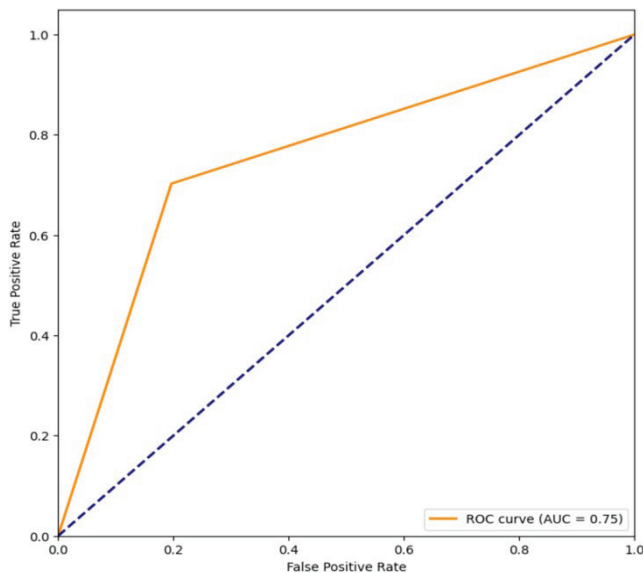


Fig. 5. Decision Tree Classifier ROC curve

V. CONCLUSION

Finally, our study on COVID-19 fake news detection with PySpark demonstrated the outstanding effectiveness of logistic regression with a 91% accuracy rate. With accuracies of 76% and 77%, respectively, decision trees and random forests performed admirably, but logistic regression's higher accuracy makes it a reliable option for differentiating between accurate and false information throughout the epidemic. This study highlights the potential influence of logistic regression on protecting public health and improving decision-making processes, highlighting its applicability as a useful tool in the current efforts to counteract misinformation surrounding COVID-19.

In the future direction, the study suggests utilizing PySpark to refine machine learning models for COVID-19 fake news detection in future research. Performance can be improved through optimization using sophisticated algorithms and hyperparameter adjustment. It is advisable to investigate the incorporation of deep learning methods, specifically neural networks, to extract complex patterns from textual data. PySpark provides streaming data processing, which can be used to construct real-time detection capabilities. It is imperative to extend the model to handle multi-modal data, integrate user feedback, and guarantee flexibility to new subjects. Prospective avenues for additional research include cross-platform analysis, a focus on explainability, examination of a variety of datasets, and cooperation with news organizations. These thoughts are meant to provide direction for continued attempts to enhance PySpark-based fake news detection within the framework of COVID-19.

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